

Lab 1B

Joseph Leone

3/21/2026

CDA3203 Computer Logic Design

Semester (Spring 2026)

Dr. Maria Petrie

Florida Atlantic University

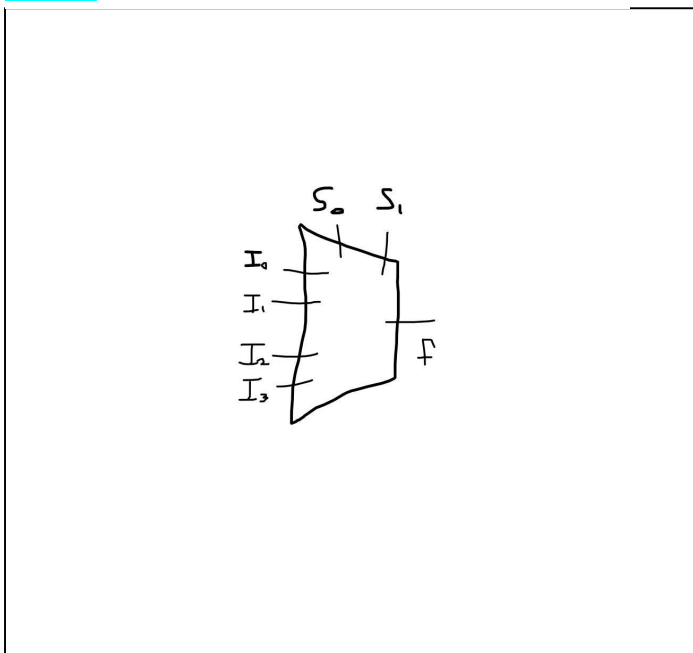
Part 2.1(50 pts total)– LAB CHECK ENTITY Create a new project in Altera Quartus in your working directory 'Lab1B_YOURNAME' using VHDL to define a 4-to-1 Mux, called Mux4to1_YourName, using the Mux2to1 component you created. YOUR DRAWING MUST MATCH YOUR CODE.

You will attend TA Hours to perform the required LAB CHECK on a COMPLETED & CORRECT 4to1 MUX Quartus Project. THIS IS REQUIRED. -25 IF NOT COMPLETED AT SUBMISSION.

TAs will add the 25 points from the lab check AFTER you have submitted this portfolio.

Dr. Petrie gives you the code for this Entity. Watch the videos!

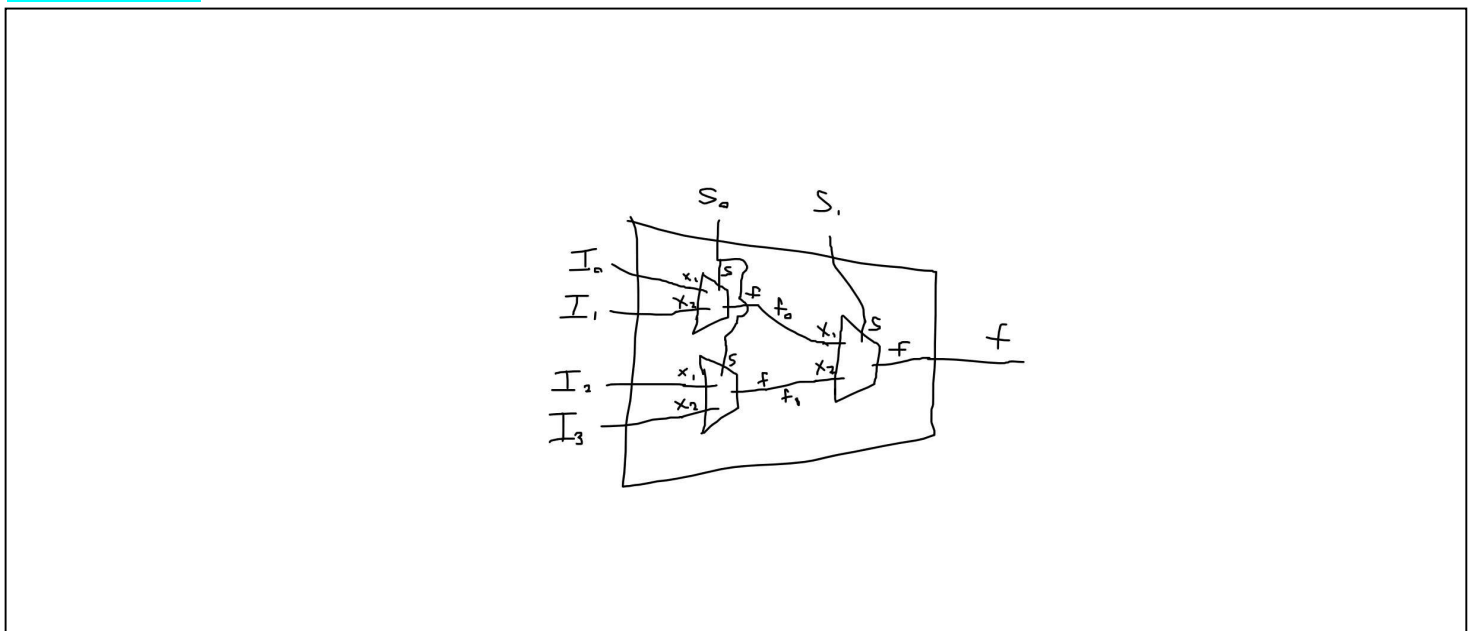
Entity Diagram (HANDDRAWN)(2pts)



Truth Table (HANDDRAWN)(1pts)

S_1	S_0	f
0	0	I_0
0	1	I_1
1	0	I_2
1	1	I_3

Architecture Diagram (HANDDRAWN)(4pts)



Project Settings(1pts)

New Project Wizard: Summary [page 5 of 5]

When you click Finish, the project will be created with the following settings:

Project directory:
G:/My Drive/Documents/Study/CDA3203/labs/Lab1B_JosephLeone/

Project name: Mux4to1_JosephLeone
Top-level design entity: Mux4to1_JosephLeone
Number of files added: 1
Number of user libraries added: 0

Device assignments:
Family name: Cyclone II
Device: EP2C35F672C6

EDA tools:
Design entry/synthesis: <None>
Simulation: <None>
Timing analysis: <None>

Operating conditions:
Core voltage: 1.2V
Junction temperature range: 0-85 °C

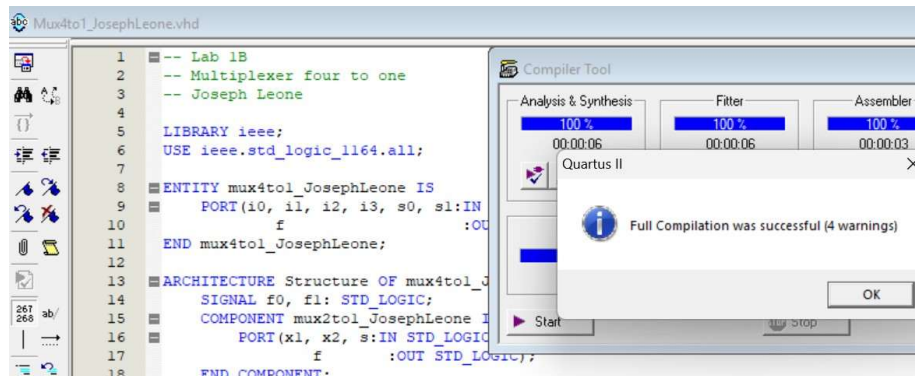
VHDL CODE(6pts)

Mux4to1_JosephLeone.vhd

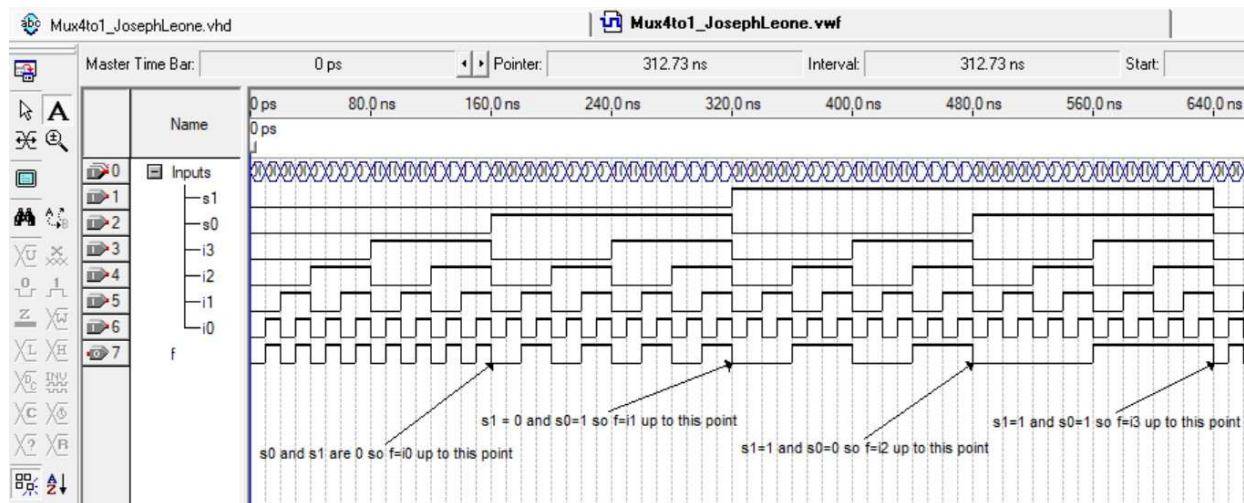
Simulator Tool

```
1  -- Lab 1B
2  -- Multiplexer four to one
3  -- Joseph Leone
4
5  LIBRARY ieee;
6  USE ieee.std_logic_1164.all;
7
8  ENTITY mux4to1_JosephLeone IS
9  PORT(i0, i1, i2, i3, s0, s1:IN STD_LOGIC;
10      f      :OUT STD_LOGIC);
11  END mux4to1_JosephLeone;
12
13  ARCHITECTURE Structure OF mux4to1_JosephLeone IS
14  SIGNAL f0, f1: STD_LOGIC;
15  COMPONENT mux2to1_JosephLeone IS
16  PORT(x1, x2, s:IN STD_LOGIC;
17      f      :OUT STD_LOGIC);
18  END COMPONENT;
19
20  BEGIN
21  mux0: mux2to1_JosephLeone PORT MAP(x1=>i0, x2=>i1, s=>s0, f=>f0);
22  mux1: mux2to1_JosephLeone PORT MAP(x1=>i2, x2=>i3, s=>s0, f=>f1);
23  mux2: mux2to1_JosephLeone PORT MAP(x1=>f0, x2=>f1, s=>s1, f=>f);
24  END Structure;
25
```

Schematics Successful Compilation(1pts)

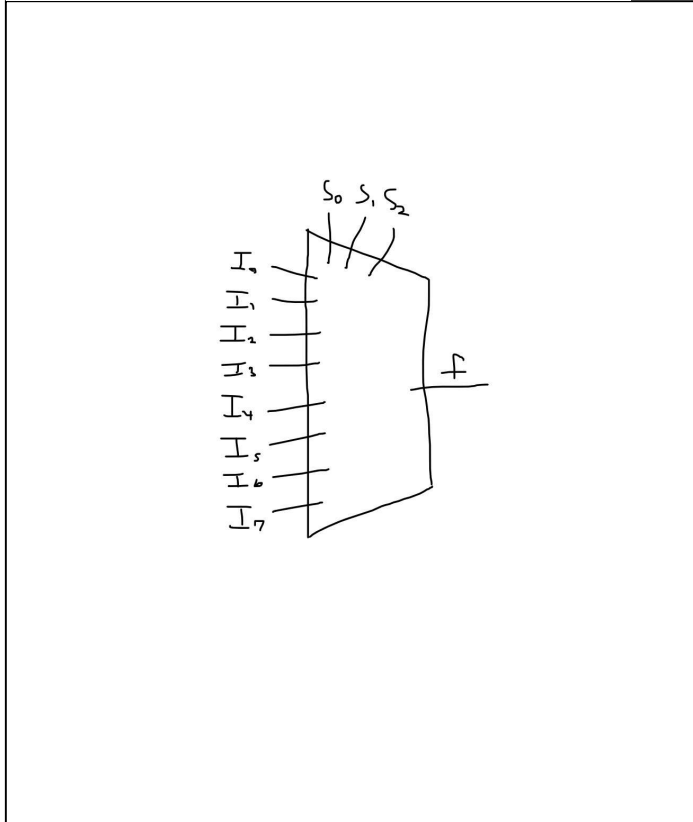


Timing Diagram (vwf)(10pts)



Part 3.2(25 pts)-Create a new project in Altera Quartus in your working directory 'Lab1B_YOURNAME' using VHDL to define an 8-to-1 Mux, called **Mux8to1_YourName**, using the Mux2to1 AND/OR Mux4to1 components you created.

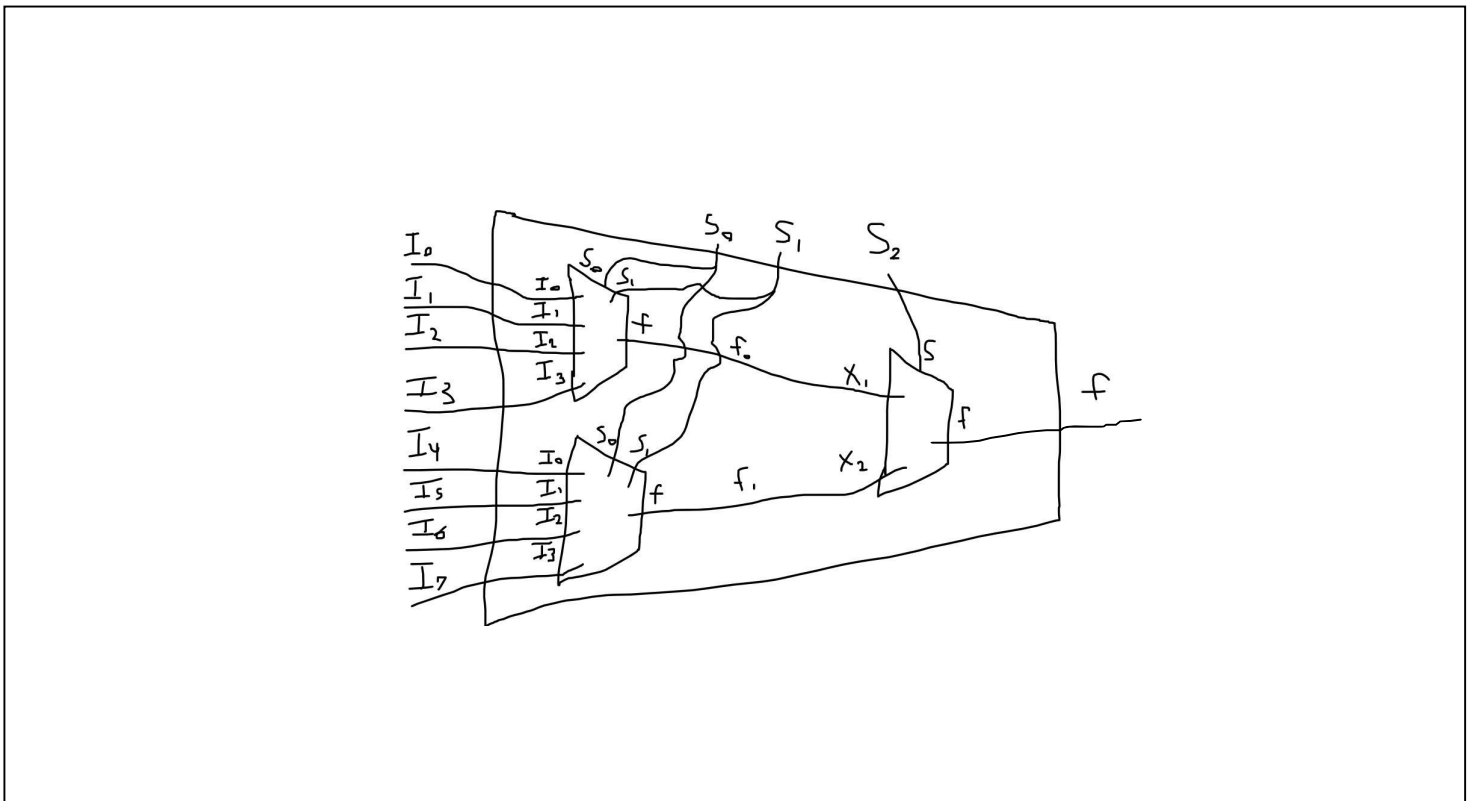
Entity Diagram (HANDDRAWN)(2pts)



Truth Table (HANDDRAWN)(1pts)

S_2	S_1	S_0	f
0	0	0	I_0
0	0	1	I_1
0	1	0	I_2
0	1	1	I_3
1	0	0	I_4
1	0	1	I_5
1	1	0	I_6
1	1	1	I_7

Architecture Diagram (HANDDRAWN)(4pts)



Project Settings(1pts)

New Project Wizard: Summary [page 5 of 5]

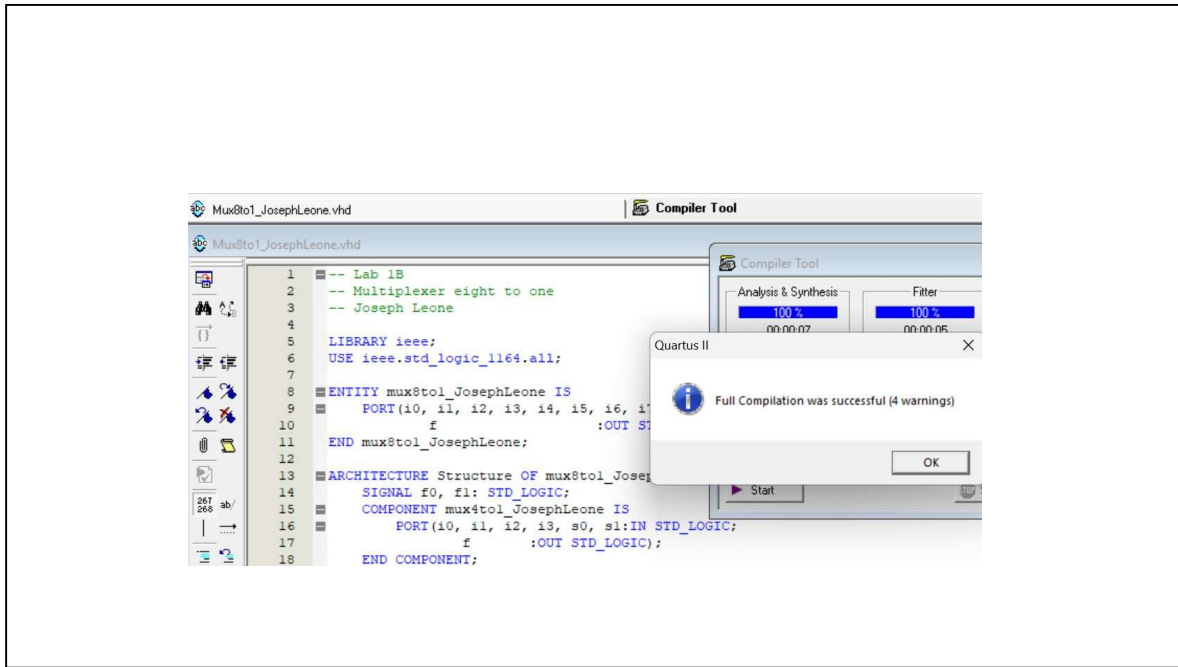
When you click Finish, the project will be created with the following settings:

Project directory:
G:/My Drive/Documents/study/cda3203/labs/Lab1B_JosephLeone/
Project name: Mux8to1_JosephLeone
Top-level design entity: Mux8to1_JosephLeone
Number of files added: 2
Number of user libraries added: 0
Device assignments:
Family name: Cyclone II
Device: EP2C35F672C6
EDA tools:
Design entry/synthesis: <None>
Simulation: <None>
Timing analysis: <None>
Operating conditions:
Core voltage: 1.2V
Junction temperature range: 0-85 °C

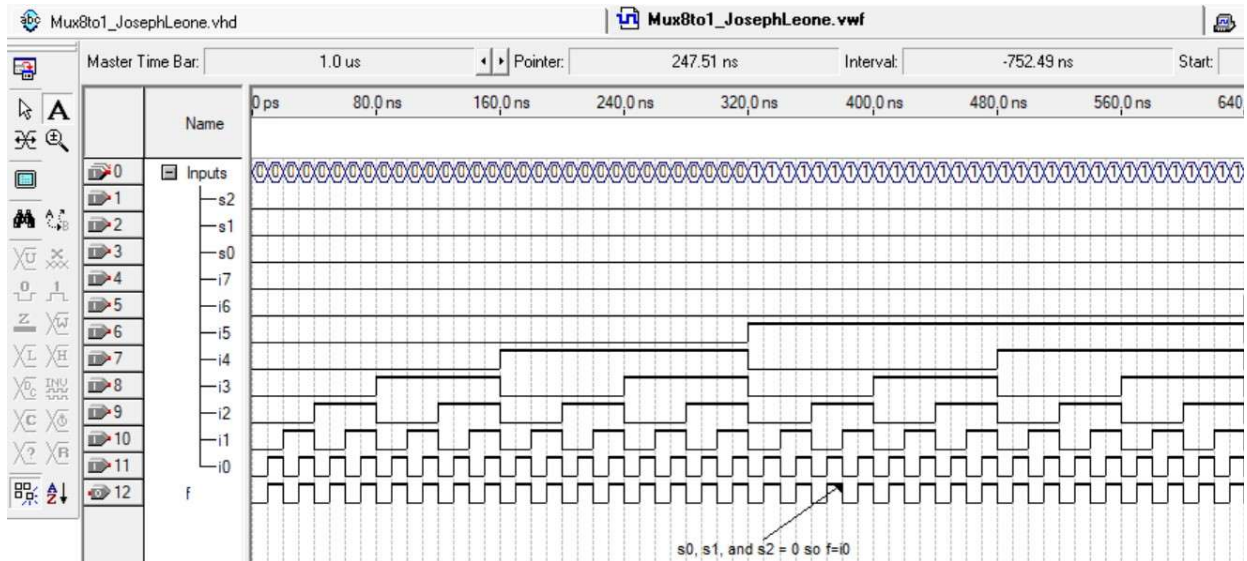
VHDL CODE(6pts)

```
Mux8to1_JosephLeone.vhd | Mux8to1_JosephLeone.vwf
1  -- Lab 1B
2  -- Multiplexer eight to one
3  -- Joseph Leone
4
5  LIBRARY ieee;
6  USE ieee.std_logic_1164.all;
7
8  ENTITY mux8to1_JosephLeone IS
9  PORT(i0, i1, i2, i3, i4, i5, i6, i7, s0, s1, s2:IN STD_LOGIC;
10      f:OUT STD_LOGIC);
11  END mux8to1_JosephLeone;
12
13  ARCHITECTURE Structure OF mux8to1_JosephLeone IS
14  SIGNAL f0, f1: STD_LOGIC;
15  COMPONENT mux4to1_JosephLeone IS
16  PORT(i0, i1, i2, i3, s0, s1:IN STD_LOGIC;
17      f:OUT STD_LOGIC);
18  END COMPONENT;
19  COMPONENT mux2to1_JosephLeone IS
20  PORT(x1, x2, s:IN STD_LOGIC;
21      f:OUT STD_LOGIC);
22  END COMPONENT;
23
24  BEGIN
25  mux0: mux4to1_JosephLeone PORT MAP(i0=>i0, i1=>i1, i2=>i2, i3=>i3, s0=>s0, s1=>s1, f=>f0);
26  mux1: mux4to1_JosephLeone PORT MAP(i0=>i4, i1=>i5, i2=>i6, i3=>i7, s0=>s0, s1=>s1, f=>f1);
27  mux2: mux2to1_JosephLeone PORT MAP(x1=>f0, x2=>f1, s=>s2, f=>f);
28  END Structure;
29
```

Schematics Successful Compilation(1pts)



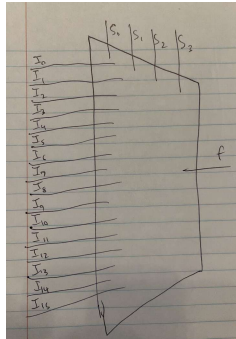
Timing Diagram (vwf)(10pts)



Part 3.3(25 pts)- Create a new project in Altera Quartus in your working directory

'Lab1B_YOURNAME' using VHDL to define a **16-to-1 Mux**, called Mux16to1_YourName, **using the Mux2to1, Mux4to1, AND/OR Mux8to1 component you created.**

Entity Diagram (HANDDRAWN)(10pts)



Truth Table (HANDDRAWN)(5pts)

S_2	S_1	S_0	f
0	0	0	I_0
0	0	1	I_1
0	0	1	I_2
0	1	0	I_4
0	1	0	I_5
0	1	1	I_6
0	1	1	I_7
1	0	0	I_8
1	0	0	I_9
1	0	1	I_{10}
1	0	1	I_{11}
1	1	0	I_{12}
1	1	0	I_{13}
1	1	1	I_{14}
1	1	1	I_{15}

Architecture Diagram (HANDDRAWN)(10pts)

